Strong convergence to the spectral edge

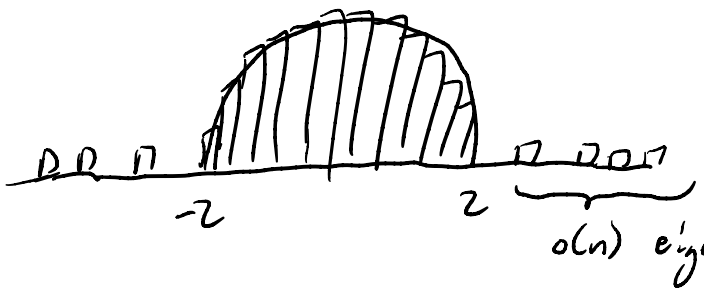
$X \in \mathbb{R}^{n \times n}$ Wigner, μ_n its ESD, μ_{sc} semicircle law on $[-2, 2]$.

Wigner's Thm shows: $\mu_n \to \mu_{sc}$ weakly a.s., i.e.

$$\mu_n([a,b]) = \frac{1}{n}|\{i : \lambda_i \in [a,b]\}| \to \int_a^b d\mu_{sc}(x) \quad \forall \text{ fixed } [a,b].$$

This implies $\liminf_{n\to\infty} \|X\|_{op} \geq 2$ a.s., but not $\lim_{n\to\infty} \|X\|_{op} = 2$ a.s.

[If $x_{ij} = \frac{1}{\sqrt{n}} z_{ij}$ where $\mathbb{E}|z_{ij}|^3 = \infty$, then:



- $\mu_n \to \mu_{sc}$ weakly a.s.
- $\limsup_{n\to\infty} \|X\|_{op} = \infty$ a.s.]

Thm: Under the conditions for X in Lecture 1 (including $\mathbb{E}|z_{ij}|^p \leq C_p \ \forall p \geq 1$), $\|X\|_{op} \to 2$ a.s. as $n \to \infty$.

Proof #1: Moment method

Idea: For any even integer $k \geq 1$

$$\|X\|_{op}^k = \max_{i=1}^n |\lambda_i|^k \leq \sum_{i=1}^n \lambda_i^k = \operatorname{Tr} X^k \leq n \cdot \|X\|_{op}^k$$

So $(\operatorname{Tr} X^k)^{1/k}$ upper bounds $\|X\|_{op}$, and is loose by factor at most $n^{1/k} = e^{\frac{\log n}{k}}$. Taking $k \gg \log n$ makes $n^{1/k} \approx 1$.

Lemma: There exist constants $a, b > 0$ s.t. if $X_{ij} = \frac{1}{\sqrt{n}} z_{ij}$ w/ $\mathbb{E} z_{ij} = 0$, $\mathbb{E} z_{ij}^2 \le 1$ $\forall i, j$, and $\mathbb{P}[|z_{ij}| \le n^a] = 1$, then for any $k = k(n) \le n^b$,

$$\mathbb{E}\, \mathrm{Tr}\, X^k \le 2n \cdot 2^k \quad \text{for all large } n.$$

Proof of Thm: Given Wigner $X = \left(\frac{1}{\sqrt{n}} z_{ij}\right)_{i,j=1}^n$, let

$$\widetilde{X} = \left(\frac{1}{\sqrt{n}} \widetilde{z}_{ij}\right)_{i,j=1}^n, \quad \widetilde{z}_{ij} = z_{ij} \mathbb{1}\{|z_{ij}| \le \tfrac{n^a}{2}\}$$

Check that:

- $\mathbb{P}[X \ne \widetilde{X}] = \mathbb{P}\left[\max_{i,j} |z_{ij}| > \frac{n^a}{2}\right] \le n^2 \frac{\max_{i,j} \mathbb{E}|z_{ij}|^p}{(n^a/2)^p} \le \frac{C_p}{n^{ap/2}}$

  Take p large enough $\Rightarrow X = \widetilde{X}$ a.s. for all large n.

- $\|\mathbb{E}\widetilde{X}\|_{op} = \left\|\left(-\frac{1}{\sqrt{n}} \mathbb{E} z_{ij} \mathbb{1}\{|z_{ij}| > \frac{n^a}{2}\}\right)_{i,j=1}^n\right\|_{op} \to 0$ as $n \to \infty$.

- $\widetilde{X} - \mathbb{E}\widetilde{X}$ satisfies conditions of Lemma.

By the Lemma: For any even $k \le n^b$,

$$\mathbb{P}[\|\widetilde{X} - \mathbb{E}\widetilde{X}\|_{op} \ge 2+\varepsilon] \le \frac{1}{(2+\varepsilon)^k} \mathbb{E}\|\widetilde{X} - \mathbb{E}\widetilde{X}\|_{op}^k$$

$$\le \frac{1}{(2+\varepsilon)^k} \mathbb{E}\, \mathrm{Tr}(\widetilde{X} - \mathbb{E}\widetilde{X})^k \le 2n\left(\frac{2}{2+\varepsilon}\right)^k.$$

Take $k \approx C \log n$, $C = C(\varepsilon)$ large enough

$\Rightarrow \|\widetilde{X} - \mathbb{E}\widetilde{X}\|_{op} < 2+\varepsilon$ a.s. for all large n

$\Rightarrow \limsup_{n\to\infty} \|\widetilde{X} - \mathbb{E}\widetilde{X}\|_{op} \le 2$ a.s. $\Rightarrow \limsup_{n\to\infty} \|X\|_{op} \le 2$ a.s. ∎

Proof of Lemma: Recall $\mathbb{E}\,\mathrm{Tr}\,X^k = \sum_{i_1,\dots,i_k} \mathbb{E}\, x_{i_1 i_2} x_{i_2 i_3} \cdots x_{i_k i_1}$.

Summands $(i_1,\dots,i_k) \leftrightarrow$ paths $i_1 \to i_2 \to \cdots \to i_k \to i_1$ on graph over $\{1,\dots,n\}$.

For a path γ, let $v(\gamma)$ = # distinct visited vertices

$e(\gamma)$ = # distinct non-self-loop edges.

Then $v(\gamma) \le e(\gamma)+1$ since γ connected

- $\mathbb{E}\, x_{i_1 i_2} \cdots x_{i_k i_1} = 0$ unless each edge walked at least twice

Let $p(\gamma)$ = # steps of γ on self-loop

$q(\gamma)$ = # steps of γ on non-self-loop traversed ≥ 3 times

$r(\gamma)$ = # steps of γ on non-self-loop traversed twice

$$\text{Then } e(\gamma) \le \frac{q(\gamma)}{3} + \frac{r(\gamma)}{2} = \frac{q(\gamma)}{3} + \frac{k - p(\gamma) - q(\gamma)}{2}$$

$$\le \frac{k}{2} - \frac{p(\gamma)+q(\gamma)}{6}$$

$$\Rightarrow p(\gamma)+q(\gamma) \le 6\left(\frac{k}{2} - e(\gamma)\right) \le 6\left(\frac{k}{2}+1-v(\gamma)\right)$$

$$\left|\mathbb{E}\, x_{i_1 i_2} \cdots x_{i_k i_1}\right| = \frac{1}{n^{k/2}} \left|\mathbb{E}\, z_{i_1 i_2} \cdots z_{i_k i_1}\right|$$

$$\le \frac{1}{n^{k/2}}\, n^{a(p(\gamma)+q(\gamma))} \times 1^{r(\gamma)} \le \frac{1}{n^{k/2}} \cdot n^{6a\left(\frac{k}{2}+1-v(\gamma)\right)}$$

- To count # paths visiting $v(\gamma)$ vertices, divide steps of γ into 3 cases:

"+": visits a new vertex not previously visited

"−": second traversal of edge (in either direction) corresponding to previous "+"

"*": all other steps.

Eg: $1\to2\to3\to4\to3\to5\to2\to1\to2\to3\to5\to2\to1$

$+\ +\ +\ -\ +\ *\ -\ *\ -\ -\ *\ *$

- The "+" steps form a tree on $v(\gamma)$ vertices

  $\Rightarrow$ exactly $v(\gamma)-1$ "+" and $v(\gamma)-1$ "−" steps,

  so $k-2(v(\gamma)-1) = 2\left(\frac{k}{2}+1-v(\gamma)\right)$ "*" steps

  $\Rightarrow$ # possible "+", "−", "*" sequences is

  $$\le \binom{k}{\#\text{ "*"}} \cdot 2^{k-\#\text{ "*"}} \le k^{2\left(\frac{k}{2}+1-v(\gamma)\right)} \cdot 2^k.$$

- Given a specific "+", "−", "*" sequence:

  (1) n choices for starting vertex i_1

  (2) If step t is "+", $\le n$ choices for next vertex i_{t+1}

  (3) If step t is "−", let A_t be the set of edges traversed by some previous "+", but not yet by "−". If A_t has unique edge incident to i_t, then only 1 choice for i_{t+1}.

Call this "$-$" step forced, all other "$-$" steps ambiguous.

(4) If step t is ambiguous "$-$" or "$*$" then

$\leq v(\gamma) \leq k$ choices for i_{t+1}.

So # paths $\leq n^{v(\gamma)} \cdot k^{\#\text{ambiguous } "-"} \cdot k^{\#\ "*"}$

- Claim: # ambiguous "$-$" $\leq$ # "$*$"

Proof: Let c_t be number of connected components in $A_t \cup \{i_t\}$. Since A_1, A_{k+1} are empty, $c_1 = c_{k+1} = 1$.

(1) Each "$+$" keeps c_t the same

(2) Each forced "$-$" keeps c_t the same.

(3) Each ambiguous "$-$" disconnects one component of $A_t \cup \{i_t\}$ into two components (otherwise A_t has a cycle, and cannot be a subset of the tree of "$+$" edges), hence $c_{t+1} = c_t + 1$.

(4) Each "$*$" does not change A_t, only $\{i_t\}$, hence $c_{t+1} \geq c_t - 1$.

Since $c_1 = c_{k+1}$, this implies # ambiguous "$-$" $\leq$ # "$*$".

Thus: # paths visiting $v(\gamma)$ vertices

$$\leq k^{2\left(\frac{k}{2}+1-v(\gamma)\right)} \cdot 2^k \cdot n^{v(\gamma)} \cdot k^{4\left(\frac{k}{2}+1-v(\gamma)\right)}$$

$$\Rightarrow \mathbb{E}\,\mathrm{Tr}\,X^k = \sum_{v=1}^{k/2+1} \sum_{\gamma : v(\gamma)=v} \mathbb{E}\, X_{i_1 i_2} \cdots X_{i_{k-1} i_1}$$

$$\le \sum_{v=1}^{k/2+1} \left[\frac{1}{n^{k/2}} \cdot n^{6a(\frac{k}{2}+1-v)} \cdot 2^k n^v k^{6(\frac{k}{2}+1-v)} \right]$$

$$\le n \cdot 2^k \cdot \sum_{v=1}^{k/2+1} \left(\frac{n^{6a} k^6}{n} \right)^{\frac{k}{2}+1-v} \le n \cdot 2^k \sum_{i=0}^{\infty} \left(\frac{n^{6a+6b}}{n} \right)^i$$

If $a+b<6$, then this is at most $2n \cdot 2^k$ for all large n. ∎

Proof #2: Resolvent method

Notation: For nonnegative r.v.'s A_n, B_n, write

$$A_n \prec B_n \iff \forall \text{ fixed } \varepsilon, D>0,\ \mathbb{P}[A_n \ge n^\varepsilon B_n] \le \frac{1}{n^D} \text{ for all large } n.$$

Let $m_n(z) = \frac{1}{n}\mathrm{Tr}\,(X - zI)^{-1}$, $m_{sc}(z)$ Stieltjes transform of semicircle law.

Lemma: Fix any $\varepsilon > 0$. There exist constants $a, b > 0$ s.t.

for any $z \equiv z_n = \lambda + i\eta$ w/ $\eta \in [n^{-a}, 1]$ and $|\lambda| \in [2+\varepsilon, n^b]$,

$$|m_n(z) - m_{sc}(z)| \prec \frac{1}{n}.$$

Proof of Thm: For any even k,

$$\mathbb{P}[\|X\|_{op} \ge n^b] \le \frac{1}{n^{bk}} \mathbb{E}\,\|X\|_{op}^k \le \frac{1}{n^{bk}} \mathbb{E}\,\mathrm{Tr}\,X^k$$

$$= \frac{1}{n^{bk}} \mathbb{E}\, X_{i_1 i_2} \cdots X_{i_k i_1} \le \frac{1}{n^{bk}} \cdot \frac{1}{n^{k/2}} \cdot C_k n^{k/2+1}.$$

Pick k large enough $\Rightarrow \|X\|_{op} \prec n^b$ a.s. for all large n.

For any $z = \lambda + i\eta$ w/ $\eta \in [n^{-a}, 1]$, $|\lambda| \in [2+\varepsilon, n^b]$:

$$\left| \eta \, \mathrm{Im}\left(m_n(z) - m_{sc}(z)\right) \right| = \left| \frac{1}{n} \sum_{i=1}^{n} \frac{\eta^2}{(\lambda_i - \lambda)^2 + \eta^2} - \int \frac{\eta^2}{(x-\lambda)^2 + \eta^2} \, d\mu_{sc}(x) \right| \lesssim \frac{\eta}{n}$$

Set $\eta = n^{-a}$ and apply this at $z = \lambda + i\sqrt{k}\eta \in \mathbb{C}$ $k = 1, \dots, K \equiv K(a)$:

$$\left| \frac{1}{n} \sum_{i=1}^{n} \frac{\eta^2}{(\lambda_i - \lambda)^2 + k\eta^2} - \int \frac{\eta^2}{(x-\lambda)^2 + k\eta^2} \, d\mu_{sc}(x) \right| \lesssim \frac{\eta}{n}$$

Note: $$\frac{\eta^2}{x^2 + k\eta^2} - \frac{\eta^2}{x^2 + (k+1)\eta^2} = \frac{\eta^4}{(x^2 + k\eta^2)(x^2 + (k+1)\eta^2)}$$

$$\frac{\eta^4}{(x^2 + k\eta^2)(x^2 + (k+1)\eta^2)} - \frac{\eta^4}{(x^2 + (k+1)\eta^2)(x^2 + (k+2)\eta^2)} = \frac{2\eta^6}{\prod_{j=k}^{k+2} (x^2 + j\eta^2)}$$

etc. $\Rightarrow$ $\dfrac{\eta^{2K}}{\prod_{j=1}^{K} (x^2 + j\eta^2)}$ is a linear combination of $\left\{ \dfrac{\eta^2}{x^2 + k\eta^2} \right\}_{k=1}^{K}$

$$\Rightarrow \left| \underbrace{\frac{1}{n} \sum_{i=1}^{n} \frac{\eta^{2K}}{\prod_{j=1}^{K} ((\lambda_i - \lambda)^2 + j\eta^2)}}_{\geq \frac{1}{n} \sum_{i=1}^{n} \mathbb{1}\{|\lambda_i - \lambda| \leq \eta\} \cdot \frac{1}{\prod_{j=1}^{K} (j+1)}} - \underbrace{\int \frac{\eta^{2K}}{\prod_{j=1}^{K} ((x-\lambda)^2 + j\eta^2)} \, d\mu_{sc}(x)}_{\leq \frac{\eta^{2K}}{\varepsilon^{2K}}, \text{ b/c } |\lambda| \geq 2+\varepsilon} \right| \lesssim \frac{\eta}{n}$$

$$\Rightarrow \left| \{ i : |\lambda_i - \lambda| \leq \eta \} \right| \lesssim n \cdot \eta^{2K} + \eta = n^{1-2Ka} + n^{-a}$$

Take K large enough s.t. $1 - 2Ka < 0$: For any $D > 0$, all large n,

$$\mathbb{P}\left[X \text{ has eigenvalue } \lambda_i \text{ s.t. } |\lambda_i - \lambda| \leq \eta \right] \leq \frac{1}{n^D}$$

Take union bound over η-spaced grid of λ s.t. $|\lambda| \in [2+\varepsilon, n^D]$:

$$\mathbb{P}[X \text{ has eig. } \lambda_i \text{ s.t. } |\lambda_i| \in [2+\varepsilon, n^D]] \le \frac{1}{n^D}$$

So $\|X\|_{op} \le 2+\varepsilon$ a.s. for all large $n \Rightarrow \limsup_{n\to\infty} \|X\|_{op} \le 2$ a.s. ∎

To prove the Lemma: Denote

$$X = \begin{pmatrix} x_{ii} & x_i^T \\ x_i & X^{(i)} \end{pmatrix}, \quad R = (X - zI)^{-1}, \quad R^{(i)} = (X^{(i)} - zI)^{-1}.$$

Index $X^{(i)}, R^{(i)}, x_i$ by $[n]\setminus\{i\}$, let $e_j = (0, \dots, 0, \underbrace{1}_{\text{index } j}, 0, \dots, 0)$

Prop (resolvent identities):

(a) $R_{ii} = \dfrac{1}{x_{ii} - z - x_i^T R^{(i)} x_i}$

(b) $\forall i \neq j$: $R_{ij} = -R_{ii} \cdot x_i^T R^{(i)} e_j$

(c) $\forall j,k \neq i$ (possibly $j=k$): $R_{jk} = R^{(i)}_{jk} + \dfrac{R_{ij} R_{ik}}{R_{ii}}$

Proof: By Schur complement: $R_{ii} = \dfrac{1}{x_{ii} - z - x_i^T R^{(i)} x_i}$ and

$$R = \begin{pmatrix} x_{ii} - z & x_i^T \\ x_i & X^{(i)} - zI \end{pmatrix}^{-1} = \begin{pmatrix} R_{ii} & -R_{ii} \cdot x_i^T R^{(i)} \\ -R_{ii} \cdot R^{(i)} x_i & R^{(i)} + R_{ii} \cdot R^{(i)} x_i x_i^T R^{(i)} \end{pmatrix}$$

Thm $R_{ij} = -R_{ii} \cdot x_i^T R^{(i)} e_j \quad \forall j \neq i$,

$$R_{jk} = R^{(i)}_{jk} + R_{ii} \cdot e_j^T R^{(i)} x_i \cdot x_i^T R^{(i)} e_k \quad \forall j,k \neq i$$

$$= R^{(i)}_{jk} + R_{ii}\left(-\frac{R_{ij}}{R_{ii}}\right)\left(-\frac{R_{ik}}{R_{ii}}\right) = R^{(i)}_{jk} + \frac{R_{ij} R_{ik}}{R_{ii}}.$$

∎

Proof of Lemma: Let $z = \lambda + i\eta$ w/ $\eta \in [n^{-a}, 1]$, $|\lambda| \in [2+\varepsilon, n^b]$.

① Preliminary bound: Let

$$\frac{1}{R_{ii}} = X_{ii} - z - X_i^T R^{(i)} X_i$$

$$= -z + \underbrace{\left(X_{ii} - X_i^T R^{(i)} X_i + \frac{1}{n}\operatorname{Tr} R^{(i)}\right)}_{Y_i} + \underbrace{\left(\frac{1}{n}\operatorname{Tr} R - \frac{1}{n}\operatorname{Tr} R^{(i)}\right)}_{Z_i} - m_n(z)$$

Recall: • $|Z_i| \le \frac{C}{n\eta}$ by eigenvalue interlacing.

• $\forall p \ge 1$:
$$\begin{cases} \mathbb{P}\left[|X_{ii}| \ge \frac{1}{\sqrt{n}} t\right] \le C_p \left(\frac{1}{t}\right)^p \\ \mathbb{P}\left[|X_i^T u| \ge \frac{1}{\sqrt{n}} t\right] \le C_p \left(\frac{\|u\|_2}{t}\right)^p \\ \mathbb{P}\left[|X_i^T A X_i - \frac{1}{n}\operatorname{Tr} A| \ge \frac{1}{n} t\right] \le C_p \left(\frac{\|A\|_F}{t}\right)^p \end{cases}$$

for any deterministic u, A (Lecture 3)

$\Rightarrow |X_{ii}| \prec \frac{1}{\sqrt{n}}$, $|X_i^T u| \prec \frac{1}{\sqrt{n}}\|u\|_2$, $|X_i^T A X_i - \frac{1}{n}\operatorname{Tr} A| \prec \frac{1}{n}\|A\|_F$.

Here $\|R^{(i)}\|_F \le \sqrt{n} \cdot \|R^{(i)}\|_{op} \le \sqrt{n} \cdot \frac{1}{\eta}$. So $|Y_i| \prec \frac{1}{\sqrt{n}\cdot\eta}$.

Thus: $\frac{1}{R_{ii}} = -z - m_n(z) + Y_i + Z_i = -z - m_n(z) + O_\prec\left(\frac{1}{\sqrt{n}\cdot\eta}\right)$

Claim: If $\frac{1}{R_{ii}} = -z - m_n(z) + O_\prec(\delta)$ $\forall i = 1, \dots, n$, where $\delta \ll \eta$ (i.e. $\delta n^\varepsilon < \eta$ for some $\varepsilon > 0$) then

- $R_{ii}, m_n(z) = \frac{1}{-z-m_n(z)} + O_z\left(\frac{\delta}{|z+m_n(z)|^2}\right)$

- $\frac{1}{m_n(z)} = -z - m_n(z) + O_z(\delta)$, $|m_n(z) - m_{sc}(z)| \geq \delta$.

Proof: Let $\frac{1}{R_{ii}} = -z - m_n(z) + \delta_i$, $|\delta_i| \geq \delta$.

Note: $|z + m_n(z)| \geq \mathrm{Im}\, z + m_n(z) \geq \mathrm{Im}\, z = \eta \gg \delta$.

$$\Rightarrow R_{ii} = \frac{1}{-z - m_n(z) + \delta_i}$$

$$= \frac{1}{-z - m_n(z)} - \frac{\delta_i}{(-z-m_n(z))(-z-m_n(z)+\delta_i)}$$

$$= \frac{1}{-z-m_n(z)} + O_z\left(\frac{\delta}{|z+m_n(z)|^2}\right)$$

$$\Rightarrow m_n(z) = \frac{1}{n}\sum_{i=1}^{n} R_{ii} = \frac{1}{-z-m_n(z)} + O_z\left(\frac{\delta}{|z+m_n(z)|^2}\right)$$

Let $m_n(z) = \frac{1}{-z-m_n(z)} + \tilde{\delta}$,

$$|\tilde{\delta}| \geq \frac{\delta}{|z+m_n(z)|^2} \geq \frac{\eta}{\delta} \cdot \frac{1}{|z+m_n(z)|} \gg \frac{1}{|z+m_n(z)|}$$

$$\Rightarrow \frac{1}{m_n(z)} = \frac{1}{\frac{1}{-z-m_n(z)} + \tilde{\delta}} = -z - m_n(z) + O_z(|\tilde{\delta}|^2 |z+m_n(z)|^2)$$

$$= -z - m_n(z) + O_z(\delta).$$

Let $\frac{1}{m_n(z)} = -\tilde{z} - m_n(\tilde{z})$, $|\tilde{z} - z| \geq \delta$. For small enough $\varepsilon > 0$:

$\operatorname{Im} \tilde{z} \geq \operatorname{Im} z - \delta \cdot n^{\varepsilon} = \eta - \delta \cdot n^{\varepsilon} > 0$, i.e. $\tilde{z} \in \mathbb{C}^+$,

w/ prob. $1 - n^{-D}$ $\forall$ all large n.

Recall: For any $\tilde{z} \in \mathbb{C}^+$, $m_{sc}(\tilde{z})$ is unique root in $\mathbb{C}^+$ of

$$\frac{1}{m} = -\tilde{z} - m. \quad \text{So } m_n(z) = m_{sc}(\tilde{z})$$

m_{sc} is $\frac{1}{\varepsilon^2}$-Lipschitz over $\{\lambda + i\eta : |\lambda| \geq 2 + \varepsilon\}$.

$$\Rightarrow |m_n(z) - m_{sc}(z)| = |m_{sc}(\tilde{z}) - m_{sc}(z)| \leq C(\varepsilon)|\tilde{z} - z| < \delta.$$

Using claim: $|m_n(z) - m_{sc}(z)| < \frac{1}{\sqrt{n}\cdot\eta}$ if $\frac{1}{\sqrt{n}\eta} \ll \eta$ (i.e. $\eta \gg n^{-1/4}$)

② Improved bound: Write $X = \sum_{k=1}^{n} \lambda_k v_k v_k^T$. Then

$$\|R e_j\|_2^2 = \left\| \sum_{k=1}^{n} \frac{1}{\lambda_k - z} \cdot v_k v_k^T e_j \right\|_2^2 = \sum_{k=1}^{n} \frac{(v_k^T e_j)^2}{|\lambda_k - z|^2}$$

$$= \frac{1}{\eta} \operatorname{Im} \sum_{k=1}^{n} \frac{(v_k^T e_j)^2}{\lambda_k - z} = \frac{1}{\eta} \operatorname{Im} R_{jj}$$

By above:

$$R_{ii}, m_n = \frac{1}{-z - m_n(z)} + O_{<}\left(\frac{1}{\sqrt{n}\cdot\eta^3}\right)$$

$$\frac{1}{-z - m_n(z)} = \frac{1}{-z - m_{sc}(z)} + O_{<}\left(\frac{1}{\sqrt{n}\cdot\eta^3}\right) = m_{sc}(z) + O_{<}\left(\frac{1}{\sqrt{n}\eta^3}\right)$$

$$\Rightarrow |R_{ii} - m_{sc}(z)| < \frac{1}{\sqrt{n}\cdot\eta^3}.$$

Recall: $m_{sc}(z) = \int_{-2}^{2} \frac{1}{x - z} d\mu_{sc}(x)$, $z = \lambda + i\eta$ w/ $|\lambda| \geq 2 + \varepsilon$

$\Rightarrow |m_{sc}(z)| \le \frac{1}{\varepsilon}$, $\operatorname{Im} m_{sc}(z) = \int_{-2}^{2} \frac{\eta}{|x-z|^2} d\mu_{sc}(x) \le \frac{\eta}{\varepsilon^2}$.

So: $|R_{ii}| \prec 1$, $|m_n(z)| \prec 1$, $|\operatorname{Im} R_{ii}| \prec \eta$, $\|R_{ij}\|_2^2 \prec 1$, $\|R\|_F^2 \prec n$

as long as $\frac{1}{\sqrt{n}\cdot\eta^3} \ll \eta$ (i.e. $\eta \gg n^{-1/8}$)

Then: $|Y_i| \prec \frac{1}{\sqrt{n}} + \frac{1}{n}\|R^{(i)}\|_F \prec \frac{1}{\sqrt{n}}$

$$|Z_i| = \left|\frac{1}{n}\operatorname{Tr} R - \frac{1}{n}\operatorname{Tr} R^{(i)}\right|$$

$$= \left|\frac{1}{n}R_{ii} + \frac{1}{n}\sum_{j\neq i} R_{jj} - R^{(i)}_{jj}\right|$$

$$\le \frac{1}{n}|R_{ii}| + \frac{1}{n}\sum_{j\neq i}\left|\frac{R_{ij}^2}{R_{ii}}\right|$$

$$= \frac{1}{n}\underbrace{|R_{ii}|}_{\prec 1} + \frac{1}{n}\sum_{j\neq i}\underbrace{|R_{ii}|}_{\prec 1}\cdot\underbrace{|x_i^T R^{(i)} e_j|^2}_{\prec \frac{1}{n}\|R^{(i)}e_j\|_2^2 \prec \frac{1}{n}} \prec \frac{1}{n}.$$

$\Rightarrow \frac{1}{R_{ii}} = -z - m_n(z) + Y_i + Z_i = -z - m_n(z) + O_\prec\left(\frac{1}{\sqrt{n}}\right)$

$\Rightarrow R_{ii}, m_n(z) = \frac{1}{-z-m_n(z)} + O_\prec\left(\frac{1}{\sqrt{n}|z+m_n(z)|^2}\right)$,

$\frac{1}{m_n(z)} = -z - m_n(z) + O_\prec\left(\frac{1}{\sqrt{n}}\right)$,

$\left|\frac{1}{R_{ii}}\right|, \left|\frac{1}{m_n(z)}\right| \prec |z+m_n(z)| \prec z, \qquad \left|\frac{1}{z+m_n(z)}\right| \prec 1.$

Lemma (fluctuation averaging): $\left|\frac{1}{n}\sum_{i=1}^n Y_i\right| \prec \frac{|z|}{n}$

(Improves over $|Y_i| \prec \frac{1}{\sqrt{n}}$ by factor of $\frac{1}{\sqrt{n}}$.)

Assuming this lemma: $\frac{1}{n}\sum_{i=1}^{n} \frac{1}{R_{ii}} = -z - m_n(z) + O_\prec\left(\frac{|z|}{n}\right)$

Also: $\frac{1}{n}\sum_{i=1}^{n} \frac{1}{R_{ii}} = \frac{1}{n}\sum_{i=1}^{n} \frac{1}{m_n} + \frac{m_n - R_{ii}}{m_n^2} + \frac{(m_n - R_{ii})^2}{m_n^2 R_{ii}}$

$$= \frac{1}{m_n} + 0 + \frac{1}{n}\sum_{i=1}^{n} \frac{(m_n - R_{ii})^2}{m_n^2 R_{ii}} = \frac{1}{m_n} + O_\prec\left(\frac{1}{n}\right)$$

$$\Rightarrow \frac{1}{m_n(z)} = -z - m_n(z) + O_\prec\left(\frac{|z|}{n}\right)$$

Suppose $\frac{|z|}{n} \ll n^{-1/8} \ll \eta$ (i.e. $\eta \in [n^{-\gamma}, 1]$, $|\lambda| \in [2-\varepsilon, n^b]$ for $a < 1/8$, $b < 7/8$)

Then $m_n(z) = m_{sc}(\tilde{z})$ for $\tilde{z} \in \mathbb{C}^+$, $|z - \tilde{z}| \prec \frac{|z|}{n}$

Use $m_{sc}(z)$ is $\frac{C}{|z|^2}$-locally-Lipschitz $\Rightarrow |m_n(z) - m_{sc}(\tilde{z})| \prec \frac{1}{n}$. ∎

Proof sketch of Lemma (fluct. avg.): Let

$$Q_i F(X) = F(X) - \mathbb{E}[F(X) \mid X^{(i)}].$$

Then • $Q_i \frac{1}{R_{ii}} = Q_i\left[x_{ii} - z + \frac{1}{n} x_i^T R^{(i)} x_i\right]$

$$= x_{ii} + \frac{1}{n} x_i^T R^{(i)} x_i - \frac{1}{n} \operatorname{Tr} R^{(i)} = Y_i.$$

• For any F, G, $\mathbb{E}[Q_i F(X) \cdot G(X^{(i)})] = 0$.

$$\mathbb{E}\left|\frac{1}{n}\sum_{i=1}^{n} Y_i\right|^2 = \frac{1}{n^2}\sum_i \mathbb{E}|Y_i|^2 + \frac{1}{n^2}\sum_{i \neq j} \mathbb{E} Y_i \overline{Y_j}$$

$$= O_\prec\left(\frac{1}{n^2}\right) + \frac{1}{n^2}\sum_{i \neq j} \mathbb{E} Y_i \overline{Y_j} \text{ since } |Y_i| \prec \frac{1}{\sqrt{n}}$$

Apply $R_{ii} = R_{ii}^{(i)} + \frac{R_{ij}^2}{R_{jj}} \Rightarrow \frac{1}{R_{ii}} = \frac{1}{R_{ii}^{(i)}} - \frac{R_{ij}^2}{R_{jj}^2 R_{ii}^{(i)}}$

$$\mathbb{E}\, Y_i \overline{Y_j} = \mathbb{E}\left(Q_i \frac{1}{R_{ii}^{(i)}} - Q_i \frac{R_{ij}^2}{R_{jj}^2 R_{ii}^{(i)}}\right)\overline{\left(Q_j \frac{1}{R_{jj}^{(j)}} - Q_j \frac{R_{ji}^2}{R_{jj}^2 R_{ii}^{(j)}}\right)}$$

$$= \mathbb{E}\, Q_i \frac{R_{ij}^2}{R_{jj}^2 R_{ii}^{(i)}} \overline{Q_j \frac{R_{ji}^2}{R_{jj}^2 R_{ii}^{(j)}}} \quad \text{(remaining terms are 0)}$$

Here $\left|\frac{R_{ij}}{R_{jj}}\right| = |X_i^T R^{(i)} e_j| \prec \frac{1}{\sqrt{n}} \|R^{(i)} e_j\|_2 \prec \frac{1}{\sqrt{n}}$, $\left|\frac{1}{R_{ii}}\right| \prec |z|$

$$\Rightarrow \left|Q_i \frac{R_{ij}^2}{R_{jj}^2 R_{ii}^{(i)}}\right| \prec \frac{|z|}{n} \Rightarrow |\mathbb{E}\, Y_i \overline{Y_j}| \prec \frac{|z|^2}{n^2}$$

$$\Rightarrow \mathbb{E}\left|\frac{1}{n}\sum_{i=1}^n Y_i\right|^2 \prec \frac{|z|^2}{n^2}$$

An extension of this argument shows: $\forall$ fixed integer $k \geq 1$,

$$\mathbb{E}\left|\frac{1}{n}\sum_{i=1}^n Y_i\right|^{2k} \prec \frac{|z|^{2k}}{n^{2k}} \qquad (**)$$

Thm: $\forall \varepsilon, D > 0$, $\mathbb{P}\left[\left|\frac{1}{n}\sum_{i=1}^n Y_i\right| \geq n^{\varepsilon}\frac{|z|}{n}\right] \leq \frac{\mathbb{E}\left|\frac{1}{n}\sum_{i=1}^n Y_i\right|^{2k}}{n^{2\varepsilon k}\left(\frac{|z|}{n}\right)^{2k}} \leq \frac{1}{n^D}$

For large enough $k = k(\varepsilon, D)$ $\Rightarrow \left|\frac{1}{n}\sum_{i=1}^n Y_i\right| \prec \frac{|z|}{n}$. ∎

Problem 7: Prove (**). [Hint: To bound $\mathbb{E}\, Y_{i_1} \cdots Y_{i_k} \overline{Y_{i_{k+1}}} \cdots \overline{Y_{i_{2k}}}$, expand $\frac{1}{R_{ij}}$ in all other indices $\{i_1, \ldots, i_{2k}\} \setminus \{i, j\}$ using resolvent identities, and think about which terms have expectation 0.]